Sample Technical Report

1. Introduction

Wireless power links have been studied for decades [1]. Recent designs reach high efficiency at moderate distances [2–4], and resonant coupling is now the usual approach (Smith et al., 2020). Jones and Lee (2019) compared several coil geometries and found that spacing dominates the result.

All measurements in this report use the same bench setup[1] and the same load. The raw data is available at https://example.com/research/papers/2021/sample-report-final-version.pdf for anyone who wants to check the numbers.

The rest of this report describes the system, presents the results and draws some conclusions. Each section is short, so the whole report can be read in a few minutes. The work was planned in March (March 2020) and finished on time.

[1] Measured at 25 °C ambient with the probe grounded at the test point.

2. System Overview

The power stage converts the input voltage to a regulated output. The main parts are shown in the block diagram below, with the input on the left and the load on the right.

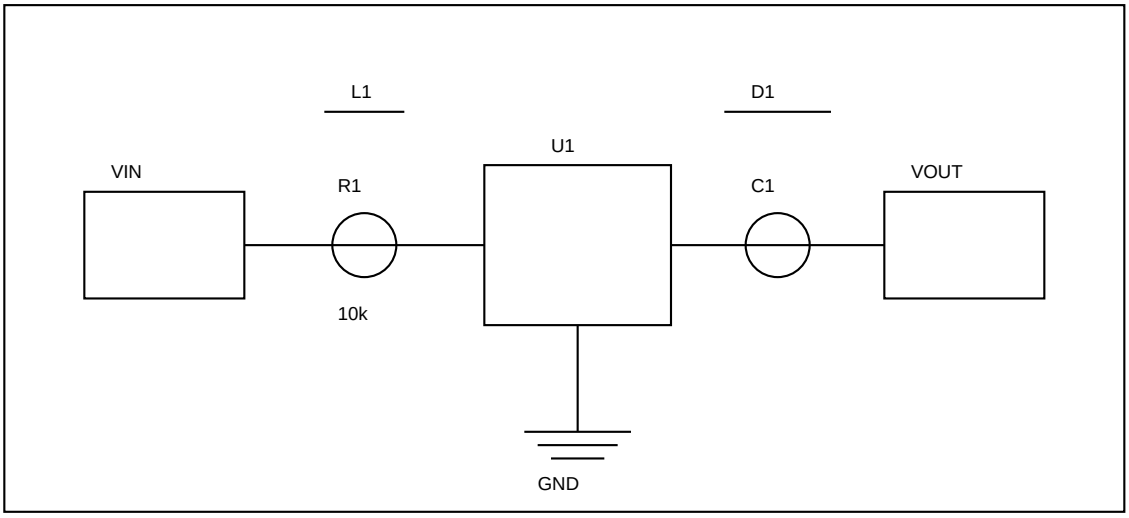

Figure 1. Block diagram of the sample power stage.

The design goals were simple and are listed here.

- Low noise
- Small size
- High efficiency
- Simple layout
- Low cost

3. Results

Table 1 summarises the gain of each stage measured on the bench.

Table 1: Stage parameters

Stage	Role	Gain
TX Stage 1	Impedance step-up	12 dB
TX Stage 2	Power amplifier	20 dB
RX Stage	Rectifier	-3 dB

Figure 1 shows that the output stage carries most of the current. The second transmit stage provides most of the gain, as expected from the simulations.

4. Conclusion

The prototype met every goal. Further work will look at thermal limits and at a smaller coil.

5. Discussion

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The right column comes second. It discusses what the results mean for the next design. The right column comes second. It discusses what the results mean for the next design.

A second paragraph in the right column closes the discussion.

References

[1] A. Smith, B. Jones, and C. Lee, “A survey of wireless power,” Journal of Power, vol. 3, pp. 1–10, 2020.

[2] D. Brown, “Resonant coupling in practice,” in Proc. Power Conf., 2018, pp. 44–50.

[3] E. Green and F. White, Coil Design, 2nd ed. New York: Example Press, 2017.

[4] G. Black, “Efficiency at distance,” IEEE Trans. Power, vol. 12, no. 4, 2021.

Appendix A. Extra Data

This appendix lists the raw readings. It should still be read aloud.